Emily Howerton

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EDUCATION AND ACADEMIC APPOINTMENTS

Princeton University, Postdoctoral Research Associate (2024 – Present)

Pennsylvania State University, Ph.D., Biology (2018 – 2023)
Dissertation: Quantitative Approaches to Improve the Management of Infectious Disease Outbreaks

College of Wooster, B.A., Mathematics and Philosophy (2013 – 2017) Summa Cum Laude
Undergraduate Thesis: On Rationality and Morality: Three Kinds of Approaches

PUBLICATIONS

Published and accepted
(* denotes co-first authors)

1. L Shandross*, **E Howerton***, L Contamin, H Hochheiser, A Krystalli, Consortium of Infectious Disease Modeling Hubs, NG Reich, EL Ray. "hubEnsembles: Ensembling methods in R". *Accepted. Statistics in Medicine*
2. BF Nielsen, SW Park, **E Howerton**, OF Lorentzen, MH Jensen, BT Grenfell. "Complex multiannual cycles of mycoplasma pneumoniae: persistence and the role of stochasticity." *Accepted*. *Proceedings of the National Academy of Sciences.*
3. SW Park, BF Nielsen, **E Howerton**, BT Grenfell, S Cobey. "Susceptible host dynamics explain pathogen resilience to perturbations." *Accepted. Proceedings of the National Academy of Sciences.*
4. **E Howerton**, TC Williams, J-S Casalegno, S Dominguez, R Gunson, K Messacar, CJE Metcalf, SW Park, C Viboud, BT Grenfell. 2025. "Using COVID-19 pandemic perturbation to model RSV-hMPV interactions and potential implications under RSV interventions". *Nature Communications* 16, 1–12. https://doi.org/10.1038/s41467-025-62358-w
5. S Loo, S-m Jung, L Contamin, **E Howerton**, S Bents, *et al.* Scenario projections of COVID-19 hospitalizations and deaths in response to immune escape levels and annual vaccination recommendations—United States, April 2024–April 2025. 2025. *JAMA Network Open* 8, e2532469. https://doi.org/10.1001/jamanetworkopen.2025.32469
6. SW Park, B Noble, **E Howerton**, BF Nielsen, SS Jiudice, L Ambroggio, S Dominguez, K Messacar, B Grenfell. 2024. "Predicting the impact of non-pharmaceutical interventions against COVID-19 on Mycoplasma pneumoniae in the United States." *Epidemics* 100808. https://doi.org/10.1016/j.epidem.2024.100808
7. **E Howerton**, TL Langkilde, and K Shea. 2024. "Misapplied management makes matters worse: Spatially explicit control leverages biotic interactions to slow invasion." *Ecological Applications*. e2974. https://doi.org/10.1002/eap.2974

8. LK Wade-Malone*, **E Howerton***, MC Runge, WJM Probert, C Viboud, and K Shea. 2024. "When do we need multiple models? Agreement between projection rank and magnitude in a multi-model setting." *Epidemics* 47: 100767. https://doi.org/10.1016/j.epidem.2024.100767
9. MC Runge, K Shea, **E Howerton**, K Yan, H Hochheiser, E Rosenstrom, WJM Probert, R Borchering, MV Marathe, B Lewis, S Venkatramanan, S Truelove, J Lessler, C Viboud. 2024. "Scenario design for infectious disease projections: Integrating Concepts from Decision Analysis and Experimental Design." *Epidemics* 100775. https://doi.org/10.1016/j.epidem.2024.100775
10. S-m Jung, S Loo, **E Howerton**, L Contamin, CP Smith, EC Carcelén, K Yan, et al. 2024. "Potential impact of annual vaccination with reformulated COVID-19 vaccines: lessons from the U.S. COVID-19 Scenario Modeling Hub." *PLOS Medicine* 21 (4): e1004387. https://doi.org/10.1371/journal.pmed.1004387
11. C Bay, G St-Onge, JT Davis, M Chinazzi, **E Howerton**, J Lessler, MC Runge, K Shea, S Truelove, C Viboud, and A Vespignani. 2024. "Ensemble^2: scenarios ensembling for communication and performance analysis." *Epidemics* 46: 100748. https://doi.org/10.1016/j.epidem.2024.100748
12. C Vanalli, **E Howerton**, F Yang, TN-A Tran, W Hu. 2024. "People and Data: Solving planetary challenges together." *Frontiers in Environmental Science* 12. https://doi.org/10.3389/fenvs.2024.1332844
13. S Loo, **E Howerton**, L Contamin, CP Smith, RK Borchering, et al. 2024. "The US COVID-19 and Influenza Scenario Modeling Hubs: Delivering long-term projections to guide policy." *Epidemics* 46:100738. https://doi.org/10.1016/j.epidem.2023.100738
14. **E Howerton**, L Contamin, LC Mullany, M Qin, et al. 2023. "Evaluation of the US COVID-19 Scenario Modeling Hub for informing pandemic response under uncertainty." *Nature Communications* 14(1):7260. https://doi.org/10.1038/s41467-023-42680-x
15. **E Howerton***, K Dahlin*, C Edholm, L Fox, M Reynolds, B Hollingsworth, G Lytle, M Walker, J Blackwood, and S Lenhart. 2023. "The effect of governance structures on optimal control of two-patch epidemic models." *Journal of Mathematical Biology* 87(5):74. https://doi.org/10.1007/s00285-023-02001-8
16. F Yang, TN-A Tran, **E Howerton**, MF Boni, JL Servadio. 2023. "Benefits of near-universal vaccination and treatment access to manage COVID-19 burden in the United States." *BMC Medicine* 21(1): 321. https://doi.org/10.1186/s12916-023-03025-z
17. K Shea, RK Borchering, WJM Probert, **E Howerton**, et al. 2023. "Multiple Models for Outbreak Decision Support in the Face of Uncertainty." *Proceedings of the National Academy of Sciences* 120 (18): e2207537120. https://doi.org/10.1073/pnas.2207537120.
18. FM Castonguay, JC Blackwood, **E Howerton**, K Shea, C Sims and JN Sanchirico. 2023. "Optimal spatial evaluation of a pro rata vaccine distribution rule for COVID-19." *Scientific Reports* 13:2194. https://doi.org/10.1038/s41598-023-28697-8
19. **E Howerton**, MC Runge, TL Bogich, RK Borchering, H Inamine, J Lessler, WJM Probert, CP Smith, S Truelove, C Viboud, and K Shea. 2023. "Context-dependent representation of within- and between-model uncertainty: aggregating probabilistic predictions in infectious disease epidemiology." Jo*urnal of the Royal Society Interface* 20:198. https://doi.org/10.1098/rsif.2022.0659

20. RK Borchering, LC Mullany, **E Howerton**, M Chinazzi, CP Smith, M Qin, NG Reich, et al. 2023. "Impact of SARS-CoV-2 Vaccination of Children Ages 5-11 Years on COVID-19 Disease Burden and Resilience to New Variants in the United States, November 2021-March 2022: A Multi-Model Study." *The Lancet Regional Health – Americas* 17(January): 100398. https://doi.org/10.1016/j.lana.2022.100398.

21. S Truelove, CP Smith, M Qin, LC Mullany, RK Borchering, J Lessler, K Shea, **E Howerton**, L Contamin, et al. 2022. "Projected resurgence of COVID-19 in the United States in July-December 2021 resulting from the increased transmissibility of the Delta variant and faltering vaccination." *eLife* 11:e73584. https://doi.org/10.7554/eLife.73584

22. **E Howerton**, MJ Ferrari, ON Bjørnstad, TL Bogich, RK Borchering, CP Jewell, JD Nichols, WJM Probert, MJ Tildesley, MC Runge, C Viboud, and K Shea. 2021. "Synergistic interventions to control COVID-19: mass testing and isolation mitigates reliance on distancing." *PLOS Computational Biology* 17 (10): e100 9518. https://doi.org/10.1371/journal.pcbi.1009518.

23. JD Nichols, TL Bogich, **E Howerton**, ON Bjørnstad, RK Borchering, MJ Ferrari, MJ Haran, CP Jewell, KM Pepin, WJM Probert, JRC Pulliam, MC Runge, MJ Tildesley, C Viboud, and K Shea. "Strategic Testing Approaches for Targeted Disease Monitoring Can Be Used to Inform Pandemic Decision-Making." *PLOS Biology* 19 (6): e3001307. https://doi.org/10.1371/journal.pbio.3001307.

    Impact: USGS Briefing to head of US Dept. of Interior and press release; reached 30K+ Facebook users; reported by 10+ news outlets

24. RK Borchering, C Viboud, **E Howerton**, CP Smith, S Truelove, MC Runge, NG Reich, et al. 2021. "Modeling of Future COVID-19 Cases, Hospitalizations, and Deaths, by Vaccination Rates and Nonpharmaceutical Intervention Scenarios — United States, April–September 2021." *Morbidity and Mortality Weekly Report* 70 (19): 719–24. https://doi.org/10.15585/mmwr.mm7019e3.

    Impact: this article is in the top 5% of all research outputs scored by Altmetric; 2-minute report by CDC Director, Dr. Rochelle Walensky, on COVID-19 White House Press Briefing (May 5, 2021); 550+ tweets; reported on 125+ news outlets

25. RD Pasteur, **E Howerton**, P Pozderac, S Young, and J Moore. (2018) "A Flight-Based Metric for Evaluating NFL Punters." *Journal of Sports Analytics* 4 (3): 201–213. https://doi.org/10.3233/JSA-180164

Submitted and in prep

1. **E Howerton**, J Lessler. "Assessing model error in counterfactual worlds." *arxiv.* https://doi.org/10.48550/arXiv.2512.00836

2. SW Park, I Holmdahl, **E Howerton**, W Yang, R Baker, GA Vecchi, S Cobey, CJE Metcalf, BT Grenfell. "Interplay between climate-driven transmission, childhood mixing, and population-level susceptibility explains a sudden shift in RSV seasonality in Japan." *medRxiv.* https://doi.org/10.1101/2025.03.02.25323095

3. L Hadley, SB Miled, HE Clapham, BA Djaafarad, L Dyson, Sebastian Funk, JR Gog, H Heesterbeek, EM Hill, **E Howerton**, et al. "The elusive 'policy maker': how understanding the diversity of perspectives and systems in governments can increase the impact of scientific research." *OSF.* https://osf.io/t3ujn/files/8xw5e

4. RN Thompson, Isaac Newton Institute Programme Participants, S Bansal, H Clapham, L Dyson, WJ Edmunds, MA Gutierrez, L Hadley, WS Hart, H Heesterbeek, TD Hollingsworth, T House, **E Howerton**, et al. “Revisiting infectious disease outbreak controllability: biological, social and public health factors.” *Submitted*
5. **E Howerton**, BT Grenfell, CJE Metcalf. “Is the honeymoon over? Dynamic impacts of abrupt reductions in vaccine uptake.” *In prep.*
6. **E Howerton***, M Chung, CJE Metcalf, B Grenfell, G Vecchi. “Why infectious disease prediction is not like weather forecasting, but can learn from it.” *In prep.*
7. H Burrows, **E Howerton**, M Chung, M Kraemer, CJE Metcalf. “Expect the unexpected: infectious disease in an era of global change.” *In prep*.
8. D Pak, **E Howerton**, WJM Probert, MC Runge, R Li, ON Bjørnstad, and K Shea. “The value of information in age-prioritization of COIVD-19 vaccination.” *In prep*.
9. SJ Bents, SL Larsen, S-m Jung, **E Howerton**, et al. “Multi-Model Insights into Racial and Ethnic Heterogeneities in Early COVID-19 Transmission and Severity in the U.S.” *In prep.*
10. D DeAngelis, J Blackwood, O Gaoue, **E Howerton**, H Inamine, O Prosper, T Ramiadantsoa, K Shea, J Yeakel, M-J Fortin, NSF Theory Group. “New ecological theory to integrate internal ecological processes and external forces.” *In prep.*
11. S Fiandrino, D Paolotti, C Bay, M Chinazzi, JT Davis, …, **E Howerton**, et al. “Adaptive multi-model ensembles for improved epidemic projections and decision support.” *In prep.*
12. Consortium of Infectious Disease Modeling Hubs, M Kerr, R Borchering, AC R, L Contamin, S Funk, H Hochheiser, **E Howerton**, A Krystalli, L Shandross, NG Reich. “Coordinating collaborative infectious disease modeling projects with the hubverse.” *In prep.*

MATERIALS FOR PUBLIC & POLICY ENGAGEMENT

1. Coauthor on Scenario Modeling Hub Reports, which are provided to the U.S. Centers for Disease Control (CDC) and other partners. Results have supported decision-making, including informing US CDC Advisory Council on Immunization Practices (ACIP) recommendations for COVID vaccination in 5-11 year old children, and multiple COVID-19 booster shots
2. Invited to publish an article in The Conversation about research evaluating US Scenario Modeling Hub impact

AWARDS

2018-2023	Inaugural Eberly College of Science Barbara McClintock Science Achievement Graduate Fellow
2023	Penn State University Alumni Association Dissertation Award
2022	Global Winner, Earnst & Young Building a Better Working World Data Challenge
2021	Peter J. Hudson Best Student Paper Award
	Jeanette Ritter Mohnkern Graduate Student Scholarship in Biology
2020	National Science Foundation Graduate Research Fellowship, Honorable Mention

2018	University Graduate Fellowship
	NCAA Post-Graduate Scholarship Recipient
2017	National Science Foundation STEM Scholar (2015-2017)
	Women's Golf Coaches Association All American Scholar (2015-2017)
	William A. Galpin Award for General Excellence in College Work
	David A. Guldin Award in Physical Education
	John F. Miller Prize in Philosophy
	William H. Wilson Price in Mathematics
2016	Edward Taylor Prize in Mathematics
2015	Andrew D. Cronin Emerging Leader Award
	Lyman C. Knight Sr. Prize in Mathematics and Physical Education

SERVICE AND LEADERSHIP

Princeton University

2025-Present	Member, *Epidemics* Editorial Advisory Board
2024-2025	Member, *Epidemics*[10] Scientific Advisory Committee
2024-Present	Member, Ecology and Evolutionary Biology Climate Committee for All
2024, 2025	Mentor and Application Review Committee Member, EEB Scholars Program

Pennsylvania State University

2020-2023	Member, Eberly College of Science Climate and Diversity Committee
2022	Project mentor, MIDAS Network/Harvard CCDD Conference to Increase Diversity in Mathematical Modeling and Public Health (including pre-conference workshop on effective teaching practices)
2020-2021	Co-president, Center for Infectious Disease Dynamics Graduate Student Association
2019	Co-designer and co-leader, Girl Scout Workshop
	Mentor, First Generation Advocates, Penn State University
	Volunteer, Discovery Space, State College, PA

GRANTS AND FUNDING

Center for Infectious Disease Dynamics Graduate Student Association Travel Grant. 2021. $1,000.

National Science Foundation Research Experience for Undergraduates (NSF-REU) Supplement to the EEID grant "US-UK Collab: Adaptive surveillance and control for endemic disease elimination" (PI: Matthew J. Ferrari). National Science Foundation. 5/1/2022-8/1/2022. $7,000. With Katriona Shea (co-PI), I advised La Keisha Wade Malone to write this proposal to fund a second summer of her research projection.

National Science Foundation Research Experience for Undergraduates (NSF-REU) Supplement to the EEID grant "US-UK Collab: Adaptive surveillance and control for endemic disease elimination" (PI: Matthew J. Ferrari). National Science Foundation. 5/1/2020-8/1/2020. $6,000. With Matthew J. Ferrari (PI) and Katriona Shea (co-PI), I helped write and was the named graduate student on this grant to mentor an undergraduate student in summer research.

Embracing uncertainty in COVID-19 management (PI: Katriona Shea). Huck Institutes of the Life Sciences Coronavirus Research Seed Fund (CRSF). 3/11/2020 - 3/10/2021. $73,785.
With advisor Katriona Shea, I co-wrote and was the named graduate student on this grant.

PRESENTATIONS

(* denotes presenter)

Invited oral presentations

1. Models of infectious disease dynamics: Making predictions to support public health response. **E Howerton***. Department of Biology and Environmental Data Science Seminar, James Madison University. April 11, 2025.
2. Multi-model scenario projections to support public health in real time: challenges and opportunities from the US Scenario Modeling Hubs. **E Howerton***. Disease Decision Analysis & Research Group, US Geological Survey. February 10, 2025.
3. Informing public health response in the face of uncertainty: Lessons from the US Scenario Modeling Hub. **E Howerton*** C Viboud, and J Lessler on behalf of the COVID-19 Scenario Modeling Hub. Isaac Newton Institute Modelling and inference for pandemic preparedness - a focussed workshop. August 5-9, 2024. Recording here
4. Synergistic interventions to control COVID-19. **E Howerton***, MJ Ferrari, ON Bjørnstad, TL Bogich, RK Borchering, CP Jewell, JD Nichols, WJM Probert, MJ Tildesley, MC Runge, C Viboud, and K Shea. Joint Mathematics Meetings. Invited oral presentation. April 6-8, 2022.
5. Synergistic interventions to control COVID-19. **E Howerton***, MJ Ferrari, ON Bjørnstad, TL Bogich, RK Borchering, CP Jewell, JD Nichols, WJM Probert, MJ Tildesley, MC Runge, C Viboud, and K Shea. Department of Biology, Pennsylvania State University. Invited oral presentation by the Jeanette Ritter Mohnkern award winner. August 24, 2021.

Oral presentations

6. The honeymoon is over: dynamic impacts of abrupt reductions in vaccine uptake. **E Howerton*,** BT Grenfell, CJE Metcalf. Epidemics^{10}. December 1, 2025.
7. Community dynamics of human respiratory pathogens: the HMPV story. **E Howerton***. Department of Ecology and Evolutionary Biology, Princeton University. February 7, 2025.
8. Modeling the impact of RSV interventions on hMPV burden and dynamics. **E Howerton*.** Fogarty International Center EPS Modeling Seminar, National Institutes of Health. October 16, 2024.
9. An introduction to Human Metapneumovirus. **E Howerton***. Life course immunity: a multi-pathogen perspective. February 9, 2024.
10. Informing pandemic response in the face of uncertainty: An evaluation of the US COVID-19 Scenario Modeling Hub. **E Howerton***, C Viboud, and J Lessler on behalf of the COVID-19 Scenario Modeling Hub. Epidemics9. November 28, 2023.
11. What does it mean to evaluate a scenario projection? Lessons from the COVID-19 Scenario Modeling Hub. **E Howerton***, J Lessler, C Viboud on behalf of the COVID-19 Scenario Modeling Hub contributors. Royal Society meeting on forecasting infectious disease incidence. March 15, 2023.
12. Collaborative open hubs for infectious disease modeling. C Viboud*, **E Howerton**, A Vespignani, J Lessler, on behalf of the COVID-19 Scenario Modeling Hub and Ebola Forecast Challenge contributors. Royal Society meeting on forecasting natural and social systems. March 14, 2023.

13. Optimizing management decisions for control of infectious disease outbreaks. **E Howerton*** Northeastern University. February 6, 2023.
14. A look back at fifteen rounds of SMH: Evaluation. **E Howerton*** COVID-19 Scenario Modeling Hub Meeting 2022 – Past, Present, and Future of Scenario Modeling for the US. September 14, 2022.
15. Looking back at one year of the COVID-19 Scenario Modeling Hub. **E Howerton***, on behalf of the COVID-19 Scenario Modeling Hub Team. MIDAS Annual Meeting. September 8, 2022.
16. Misapplied management makes matters worse: Spatially explicit control strategies that leverage biotic interactions may more effectively prevent invader spread. **E Howerton***, T Langkilde, and K Shea. August 17, 2022.
17. How to get the best out of multiple disease models. **E Howerton***, MC Runge, TL Bogich, RK Borchering, H Inamine, J Lessler, LC Mullany, WJM Probert, CP Smith, S Truelove, C Viboud, K Shea. Ecology and Evolution of Infectious Diseases. June 6-9, 2022. Recording here
18. Scenario Modeling Hub round 13 preview and evaluation. J Lessler* and **E Howerton*** on behalf of COVID-19 Scenario Modeling Hub. United States Centers for Disease Control Council of State and Territorial Epidemiologists. April 5, 2022.
19. Comparing aggregation methods for COVID-19 Scenario Modeling Hub. **E Howerton*** COVID-19 Scenario Modeling Hub Weekly Meeting. June 18, 2021.

Poster presentations

20. Cross-protection and the lags between RSV and Human Metapneumovirus outbreaks. **E Howerton***, TC Williams, J-S Casalegno, CJE Metcalf, SW Park, BT Grenfell. Ecology and Evolution of Infectious Disease Dynamics. June 25, 2024.
21. Informing pandemic response in the face of uncertainty: An evaluation of the US COVID-19 Scenario Modeling Hub. **E Howerton***, K Shea, C Viboud, and J Lessler on behalf of the COVID-19 Scenario Modeling Hub. Ecology and Evolution of Infectious Disease Dynamics. May 23, 2023.
22. Theory of combining probabilistic projections with applications in epidemiology. **E Howerton***, MC Runge, TL Bogich, RK Borchering, H Inamine, J Lessler, LC Mullany, WJM Probert, CP Smith, S Truelove, C Viboud, K Shea. Epidemics8. Poster. November 29 - December1, 2021.
23. Synergistic interventions to control COVID-19: mass testing and isolation mitigates reliance on distancing. **E Howerton***, MJ Ferrari, ON Bjørnstad, TL Bogich, RK Borchering, CP Jewell, JD Nichols, WJM Probert, MJ Tildesley, MC Runge, C Viboud, and K Shea. Ecology and Evolution of Infectious Diseases. Poster. June 14-17, 2021.

TEACHING

Pennsylvania State University

TA	Ecological and Environmental Problem Solving	Spring 2020, Spring 2021

College of Wooster

TA	First Year Seminar	Fall 2014, Fall 205
	Calculus with Algebra A	Fall 2014, Fall 2015, Fall 2016
	Calculus with Algebra B	Spring 2015, Spring 2016, Spring 2017
	Logic & Philosophy	Spring 2015
Tutor	Math Center (Pre-calculus – Calculus III)	Spring 2015 – Spring 2017

MENTORSHIP

Emily Zhang (Fall 2025 – present), Catalina Posada (Spring 2024 – Fall 2024); La Keisha Wade-Malone, NSF-REU funding (Summer 2021 – Fall 2022)

WORKSHOP PARTICIPATION

Isaac Newton Institute "Modelling and inference for pandemic preparedness - a focused workshop". University of Cambridge. August 5-9, 2024. December 8-19, 2025. Invited and funded participant.

Communication Arts & Sciences Summer Symposium "Viral Movements". Pennsylvania State University. May 15-16, 2024. Invited and funded participant.

The Royal Society Satellite Meeting "Forecasting infectious disease incidence for public health". Royal Society, London, UK. March 15-16, 2023. Invited and funded participant.

Ecology and Evolution of Infectious Disease Dynamics Workshop "Pandemic Scenario Modeling and Science Communication". June 3-6, 2022. Invited and funded participant.

American Mathematical Society "Dynamics of Infectious Diseases: Ecological Models Across Multiple Scales" Mathematics Research Community. May 30-June 5, 2021. Invited and funded participant.

National Science Foundation "Advancing Ecological Theory" workshop. Pennsylvania State University.
October 23-25, 2019.

PROFESSIONAL MEMBERSHIPS

Ecological Society of America, American Association for the Advancement of Science, American Mathematical Society, Phi Beta Kappa Academic Honor Society

REVIEWING ACTIVITY

American Journal of Epidemiology, BMC Global Health, BMC Medical Information and Decision Making, BMC Public Health, Communications Biology, eLife, Epidemics, Epidemiology and Infection, Journal of Theoretical Biology, Lancet Digital Health, Lancet Regional Health – Americas, Mathematical Biosciences, Mathematical Biosciences and Engineering, PLOS Computational Biology, PLOS One, Royal Society Interface, Royal Society Open Science, Science Advances, Scientific Reports